

SUJAL CHAVAN

Robotics & Embedded Systems Engineer

Mumbai, India | +91 7208656815 | sujalc286@gmail.com

[linkedin.com/in/sujal-chavan](https://www.linkedin.com/in/sujal-chavan) | github.com/sujal-286

Summary

Electronics and Telecommunication Engineering student specializing in robotics, embedded systems, and unmanned systems. Experienced in microcontroller-based hardware design, sensor integration, firmware development, and structured hardware testing across autonomous and IoT platforms. Seeking a Robotics/Embedded Systems internship to contribute hardware-first engineering skills to real-world autonomous systems.

Education

Mumbai University – B.Tech, Electronics & Telecommunication Engineering
CGPA: 8.56/10 (Semesters I–VI)

2023–2027

Internship Experience

PNT Robotics – Robotics Intern

On-Site, 1 Jun – 20 Jul 2026

- Contributed to hardware and firmware development of a microcontroller-based GPS and camera tracking unit for real-time personnel and asset location monitoring.
- Debugged hardware-firmware integration issues, including communication-line conflicts and power-sequencing faults affecting module reliability.
- Performed structured hardware bring-up and system-level testing to validate positioning, connectivity, and camera capture functionality.
- Collaborated with the team to align hardware design decisions with backend data and reporting requirements.

Projects

Autonomous Unmanned Aerial Vehicle (UAV)

- Built and flight-tested multi-rotor platforms, integrating flight controllers with custom power distribution networks.
- Tuned control loops and configured flight firmware parameters, implementing safety fail-safes for stable autonomous flight.

Browser-Controlled Robot

- Assembled a microcontroller-based robotic rover with GPIO-driven motor control for remote locomotion.
- Built a low-latency wireless control link with live camera feedback, tested for reliability over the wireless link.

Cellular GPS Tracking System

- Designed a cellular-based GPS tracking unit, resolving power-sequencing and communication-line conflicts during integration.
- Integrated a camera module and display for on-device image capture; validated end-to-end operation through structured bring-up testing.

Multi-Sensor Tampering Detection System

- Designed a multi-sensor system combining motion, tilt, and obstruction sensing to reduce false alarms versus single-sensor approaches.
- Implemented local alerting and real-time event notification to a monitoring dashboard, with event logging for analysis.

Core Skills

- Embedded Systems:** Embedded C/C++, Firmware Development, Sensor Integration
- Robotics:** UAV/Rover Assembly, Flight Controller Configuration, PID Tuning, Control Systems
- Hardware:** Hardware Bring-up, Power Distribution, Circuit-Level Debugging
- Communication Interfaces:** UART, I2C, SPI, PWM, GPIO, Cellular/GNSS, Wi-Fi
- Programming:** C++, Python, Java, Arduino Framework
- Development Platforms:** Microcontrollers, Single-Board Computers, Flight Controllers
- Practices:** Rapid Prototyping, Hardware/Firmware Testing & Debugging, Technical Documentation

Certifications

Introduction to Industry 4.0 and Industrial IoT – IIT Kharagpur (NPTEL), Elite Certificate, Score 77%
Certificate ID: NPTEL25CS146S1063702128